Week 2 Session B

Thursday, Jan 22 2026

Overview

- Review: LLMs
- How does a LLM work?
  - The two phases of a LLM
  - Pipelines of the two phases
- Week 2 assignment: how to create a clip?
  - Platform
  - Script

Review: LLMs

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How does an LLM work? I

- The two phases of an LLM
  - **Phase A: Training**
    Example: GPT3
    How many days roughly did it take to train the model once? **A.** 3 days **B.** 34 days **C.** 134 days
    How much did it cost to train the model each time?
    A. $300k **B.** 2.8 million **C.** 4.6 million

  - **Phase B: Inference**
    This happens everytime you use an application like ChatGPT, the model responds to you in real time.

How does an LLM work? I

Judge the following statements:

- When I chat with a LLM like GPT4o, the LLM learns new facts from our conversations.
- LLMs do not learn new facts while chatting with human, they use the patterns they learned during Phase A and make predictions.
- LLMs remember what I typed, and will update the parameters in the model as our conversation continues.
- LLMs only have limited short term memory, and this memory gets wiped when the chat resets.

How does an LLM work? II

- Pipeline that happens at **Phase 1 Training**
  - Input (all the text from the world)
  - Tokenization:
    - Token: A token is a small piece of text that the AI reads. It can be part of a word or a symbol. [From Teddy]
  - Neural network
  - Prediction (next token guess)
  - Compare guess to the real next token
  - Adjust parameters (learning!)
  - Repeat millions/billions of times

How does an LLM work? II

- What is a token/tokenization?

https://platform.openai.com/tokenizer

How does an LLM work? II

Why Do we Need Tokens from SubWords?

TheAiEdge.io

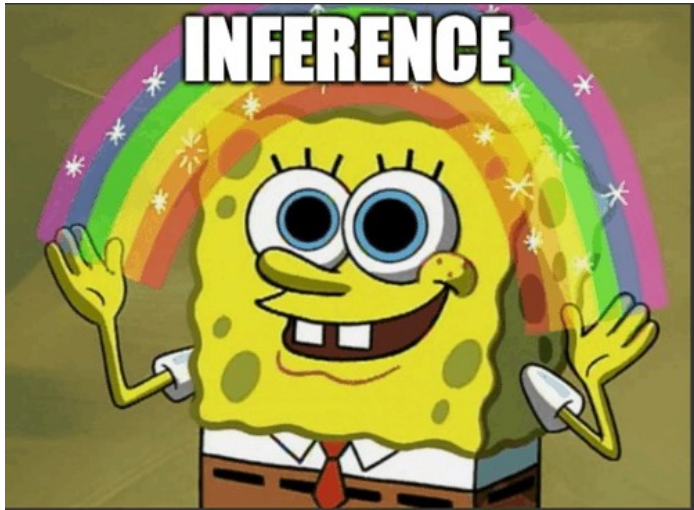
Break words into sub-words

Supercalifragilisticexpialidocious

Tokenized with
GPT-2 tokenizer

How does an LLM work? II

- Pipeline that happens at **Phase 2 Inference**
  - Your input (prompt)
  - Tokenization
  - Neural network (with frozen parameters)
  - Prediction (guess next token)
  - Prediction (guess next token)
  - Prediction (guess next token)
  - …
  - Output an answer to user



How does an LLM work? III

- Why are LLMs so powerful?

Example:
GPT2, training data:
https://huggingface.co/openai-community/gpt2
https://github.com/openai/gpt-2/blob/master/domains.txt

How does an LLM work? III

- Why are LLMs also very limited?

  Examples:
  https://huggingface.co/openai-community/gpt2
  See section “limitations and bias”

Before we move on

- Some takeaways:
  - LLMs might sound intelligent because the pattern they learned are intelligent. They don’t understand. They predict based on patterns they learned.
  - LLMs follows your prompt/input!!!!!. You give a clear prompt, it gives better output.
  - LLMs themselves are not search engines, they generate (gen AI!). They don’t retrieve information.

Week 2 Assignment

- Explain LLMs in different ways: elevator talk (to your future employers), Instagram reels/YouTube shorts (to your peer), “explain at family Thanksgiving party” (to your family members)
  - Pick a format (you just need to pick one)
  - Decide which aspect of LLMs do you want to focus on
  - Update your google doc with a complete written draft.
  - Submit a recording of your talk (video).

References

Li, Chuan. OpenAI's GPT-3 Language Model: A Technical Overview: https://lambda.ai/blog/demystifying-gpt-3